

Business Entity Resolution

Amazon ML Challenge 2026 | Team Vulcans | 27 September 2026

Problem and design goal

For each deduplicated Source 1 business, predict all matching Source 2 and Source 3 IDs, including an empty list for singletons. Training has 2.21 million Source 1 records, 10.32 million targets, and 7.64 million labeled links. Test adds France with no French training labels. The metric is macro F0.5 over Source 1 groups, so false merges and missed singletons matter alongside retrieval recall.

1. Retrieve a bounded, complementary candidate set

A local DuckDB index stores Unicode name and address tokens, target frequencies, and compact legal-suffix name keys. Each query uses rare name and address token postings within the same country. Separate name and address Jaro-Winkler top-100 quotas protect matches when either field is corrupted or absent. Compact and accent-folded name equality rescue punctuation and French accents. For India, an additional top-50 address-token-overlap channel helps cross-script names. Country is an open-set string, so France is processed without hard-coded exclusion.

2. Score pairs with a locally trained model

A LightGBM classifier trained on 3.95 million retrieved pairs from a seeded 20,000-query training sample uses 27 features: Unicode-normalized name and address similarities, token overlap, source/missingness signals, and four soft first-address-number features. It uses only supplied data; no external business lookup or remote inference. The score threshold is 0.75, with a development-selected India override of 0.65.

3. Resolve group and ownership conflicts

Rank selected links by model probability and retain at most 11 per Source 1, the largest true group in the full training labels. Assign a target predicted by multiple Source 1 records to its highest-scoring owner. Both passes preserve the exact candidate subset contract. A fixed address-number veto was rejected for this model because it reduced labeled F0.5.

Retrieval checks that changed the design

- The core candidate route reached 91.18% link recall on all 22,133 development queries with 4.52 million pairs; its oracle macro F0.5 ceiling was 0.9617.
- India address-overlap top-50 added 686 true links for 171,364 extra pairs and raised cross-script recall from 62.53% to 70.76% versus the core route.
- Accent-folded core-name equality independently added 328 true links for 53,138 extra pairs. These two rescue routes were measured separately before union.

Measured evidence and submission

All scores below are local macro F0.5; public and private leaderboard scores must be measured separately.

Held-out evaluation

Check	Scope	Result
Combined candidate oracle	22,133 development queries	0.96793 upper bound
Original 23-feature baseline	Full development, 3 output passes	0.88331
27-feature model, uniform 0.75	Full development, cap + owner	0.89574
27-feature model, India 0.65	Full development, cap + owner	0.89761
27-feature model, India 0.65	Frozen 2,000-query validation	0.89272

What the measurements mean

- The candidate oracle assumes perfect scoring on retrieved pairs. It measures the retrieval ceiling, not an achieved model score.
- The India override improved frozen validation from 0.89091 to 0.89272 and full development from 0.89574 to 0.89761; it also added false links and reduced singleton accuracy. The challenge metric favored it on both checks.
- France has no truth labels. In a 20,000-query test slice, a generic-name collision tail remained after scoring, so the top-11 cap is retained. No French F0.5 is claimed.

Reproducibility and fair play

The final ZIP contains the exact last-stage candidate TSV, matching TSV, runnable Python source, both locally trained model weights required by the pipeline, pinned dependencies, and the full methodology. The candidate TSV lists every pair fed to the final scoring model, and every predicted ID is a member of that row's candidate set. Streaming and official validators are run before submission. No supplied record text is sent to a remote service.

Team

Yash Kailas Doke, Harsh Jitendra Jain, Ayush Tiwari, and Vedant Kaulgekar.